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An Analytical Approach to Optimize Shut-In Time Based on Oil Saturation Rebalancing for Production Performance Improvement

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Abstract

To address the challenges of inconsistent shut-in performance, unclear effectiveness, and uncertain optimal duration in fractured reservoirs, this study develops an integrated numerical model coupling fracture closure, oil-water imbibition, and fracturing fluid retention during the fracturing-shut-in-production sequence. Model validation confirms its reliability in characterizing fluid migration and determining optimal shut-in duration. Key findings include: (1) Shut-in fluid migration exhibits temporal stage characteristics—fracture closure, imbibition replacement, and energy equilibrium—spatially correlated with fracture complexity; (2) While extended shut-in (30–45 days) enhances initial production through capillary imbibition, prolonged fracturing fluid retention exacerbates oil-phase permeability damage, necessitating balanced duration optimization based on cumulative oil gain; (3) Incorporating operational constraints, the "fracturing + shut-in duration" metric is proposed to reduce well-to-well variability while improving time efficiency. This work

establishes an evaluation methodology and workflow for post-fracturing shut-in optimization in volume-fractured horizontal wells.

Technical Background

Horizontal wells typically undergo a shut-in period post-fracturing prior to production, driven by both operational requirements (e.g., pipeline installation) and reservoir considerations (energy replenishment and fluid exchange). Field observations demonstrate that shut-in accelerates production onset but yields inconsistent productivity gains across wells. While extensive research has been conducted on shut-in mechanisms, debate persists regarding its necessity and optimal duration. Proponents argue that shut-in enhances early production by allowing: (1) pressure buildup near fractures through fluid retention, and (2) oil displacement into fractures via imbibition (gravity/capillary/chemical/osmotic forces), reducing water cut during flowback. Conversely, critics highlight potential drawbacks: (1) fracturing fluid invasion forms water blocks that impair oil permeability and increase pressure depletion, and (2) water-rock reactions in sensitive formations may cause formation damage. Thus, while shut-in benefits early productivity in shale oil wells, it may accelerate reservoir depletion. Current practices lack scientific guidelines for evaluating shut-in effectiveness and duration. Integrating geological, completion, microseismic, and pressure data with geoenvironmental simulations can model the fracturing-shut-in-production sequence to characterize fluid redistribution and optimize shut-in time for productivity enhancement.

Geographic Location

The Lunnan Oilfield is located on the northern edge of the Taklamakan Desert, approximately 35 km south of Luntai County in Xinjiang Uygur Autonomous Region. Benefiting from the development of the Tarim petroleum industry, the area now enjoys convenient transportation, with National Highway 314 (Wuqia–Kashgar Highway) and the Southern Xinjiang Railway serving as main transport routes along the basin's northern margin. Within the oilfield, a complete road network connects to Highway 314 at multiple points, and the northern terminus of the desert highway, which traverses the Tarim Basin, is located inside the field. The region has a small but stable migrant population, an established power grid, and reliable communications. The terrain is flat, with an average elevation of about 930 m, and vegetation is sparse, limited to patches of tamarisk and desert sedge. The climate is characterized by strong winds, low rainfall, and distinct seasonal changes, with large diurnal temperature variations. The annual mean temperature is 10.6 °C, with July averaging 26.1 °C (maximum 40 °C) and January averaging −8.1 °C (minimum −28.1 °C). Average annual precipitation is only 50.4 mm, while evaporation reaches 2,788.2 mm. The reservoir exhibits medium porosity and permeability, with relatively weak planar heterogeneity compare with T12 formation zone (Figures 1).

The Lunnan 2 TI oil-bearing formation is characterized by fining-upward depositional cycles, with distinct sedimentary structures developing at different parts of each cycle. In the lower cycle, common features include channel-base scour surfaces and parallel bedding, often accompanied by graded bedding, cross-bedding, trough cross-bedding, and massive structures. The middle to upper cycle typically exhibits small- to micro-scale cross-bedding, trough cross-bedding, ripple lamination, and horizontal bedding, with occasional plant debris fossils, coal streaks, and bitumen in the sandstone. Abundant channel-base scour surfaces indicate a predominantly fluvial depositional environment, while parallel bedding reflects high-energy conditions often associated with scour surfaces. Graded bedding suggests gravity sorting, producing the characteristic fining-upward sequence. Massive bedding reflects rapid deposition under abruptly reduced flow energy. Cross-bedding and trough cross-bedding in the middle to upper sections indicate traction current deposition under moderate flow. Fine-grained ripple laminations near the cycle top suggest lacustrine influence under very low energy, and horizontal bedding typically reflects deposition in stagnant water. Plant debris and coal streaks are interpreted as material transported by river channels.

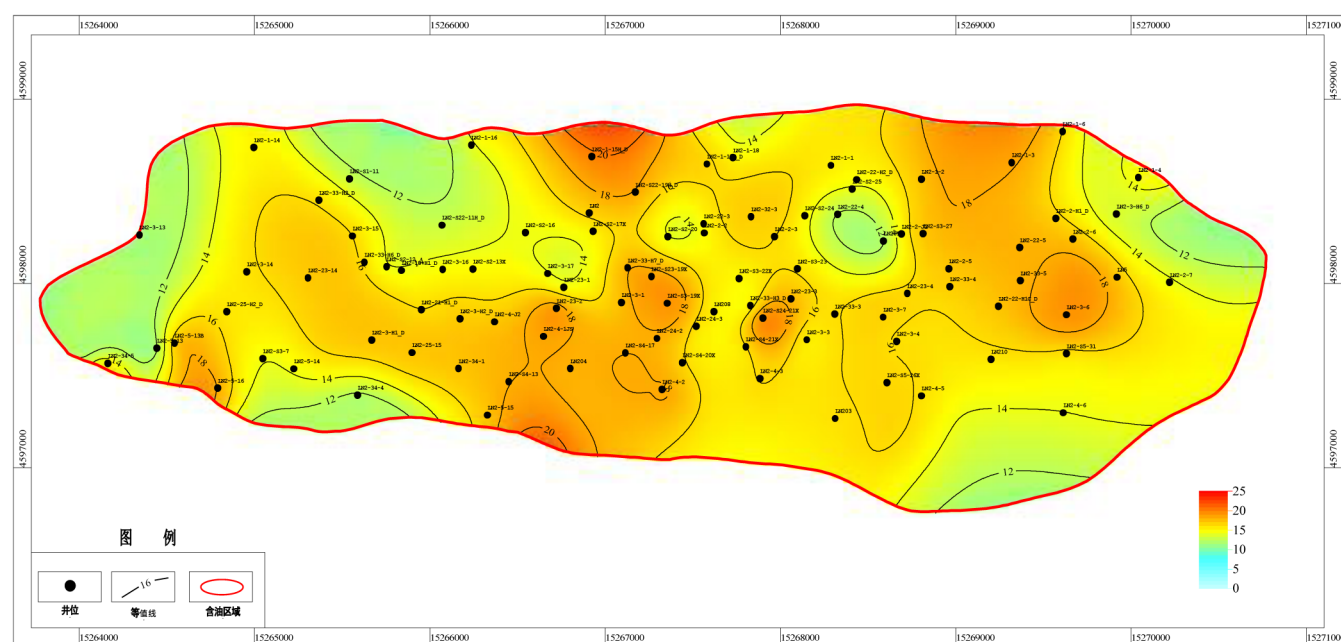


Figure 1—Porosity distribution of T12 sub-layer in Lunnan

Numerical Simulation Model Development and Case Validation

Model Development

To accurately represent the shut-in stimulation mechanism and its long-term production impact, an integrated geology–engineering software was used to simulate oil–water distribution dynamics during the fracturing–shut-in–production process. Due to limitations in existing simulation methods, which cannot simultaneously capture fracture propagation mechanics and multiphase flow behavior, the modeling workflow was divided into four steps: (1) define the study area and construct the geological model using geological and well-logging data; (2) input fracturing completion parameters to simulate fracture propagation and characterize the complex hydraulic fracture network; (3) apply unstructured grid generation to incorporate fracture geometry, dimensions, and conductivity into a dual-media model, and input high-pressure reservoir–fluid properties, relative permeability, capillary pressure, and pressure sensitivity; (4) simulate fracturing fluid injection at the actual scale to obtain pressure and saturation distributions at the end of fracturing, providing a reliable basis for subsequent shut-in and production simulations. Step 2 provides key fracture network parameters, while Step 4 reproduces the fracturing stimulation effect; in practical applications, fracture monitoring and pressure history data are used for calibration.

A single-stage fractured horizontal well in the Lunnan block was selected as a case study. To ensure computational efficiency and convergence, the model was vertically divided into three layers (upper barrier–oil-bearing layer–lower barrier) with dimensions of 460 m × 500 m × 60 m. Geological and completion parameters from the block were used to complete the fracture network model according to Steps 1 and 2. Based on fracture simulation results, propped fractures with conductivity greater than $0.1 \times 10^{-3} \mu\text{m}^2 \cdot \text{m}$ were defined as main fractures, while unpropped fractures with conductivity of $0.1 \times 10^{-3} \mu\text{m}^2 \cdot \text{m}$ were defined as secondary fractures. Different relative permeability curves were assigned to the matrix, main fractures, and secondary fractures to reflect their distinct flow capacities; matrix curves were based on core experiments, main fracture curves considered only gravity effects, and secondary fracture curves were obtained from LI et al.'s method. Capillary pressure curves, which control fracturing fluid distribution, were assigned to the matrix and secondary fractures, while main fractures ignored capillary pressure due to high permeability, with values calculated using GDANSKI's empirical formula:

$$P_c = 6.895 \times 10^{-3} \frac{\sigma}{a_2(S_w)} a_1 \left(\frac{\phi}{k} \right)^{a_3} p_c S_w \quad (1)$$

In the formula, p_c is the capillary pressure of natural fractures (MPa), σ is the interfacial tension (N/m) with a value of 0.072 N/m, S_w is the water saturation, ϕ is the porosity, k is the permeability $10^{-3} \mu\text{m}^2$, and a_1 , a_2 , and a_3 are constants with values of 1.86, 6.42, and 0.50, respectively.

Table 1—Basic Reservoir Input Parameters

Parameter	Upper Barrier	Oil-Bearing Layer	Lower Barrier
Formation pressure (MPa)	37	37	37
Formation thickness (m)	20	20	20
Porosity (%)	3	11	3
Permeability $10^{-3} \mu\text{m}^2$	0.001	0.01	0.001
Initial water saturation (%)	70	20	70
Minimum horizontal stress (MPa)	54	52	56
Maximum horizontal stress (MPa)	60	58	62
Young's modulus (GPa)	40	27	40
Poisson's ratio	0.35	0.25	0.35
Tensile strength (MPa)	3.45	3.45	3.45
Natural fracture density (MPa)	0.15	0.15	0.15

Table 2—Fracturing Completion Input Parameters

Parameter	Value	Parameter	Value
Cluster spacing (m)	10	Pumping rate (m^3/min)	14
Slickwater viscosity ($\text{mPa}\cdot\text{s}$)	5	Total fracturing fluid volume (m^3)	1300
Crosslinked gel viscosity ($\text{mPa}\cdot\text{s}$)	200	Slickwater proportion (%)	40

To realistically simulate the fracture system's expansion during the fracturing injection stage and subsequent compaction during the shut-in stage, the exponential empirical formula proposed by LIAO et al. was applied to characterize the pressure sensitivity of fracture porosity and permeability:

$$\begin{cases} \frac{\phi_f}{\phi_{f0}} = e^{C_f P_{net}} \\ \frac{k_f}{k_{f0}} = 10^{m P_{net}} \end{cases} \quad (2)$$

where ϕ_f is the fracture porosity, ϕ_{f0} is the initial fracture porosity, C_f is the fracture compressibility MPa^{-1} , P_{net} is the net fracture pressure (MPa), k_f is the fracture permeability $10^{-3} \mu\text{m}^2$, k_{f0} is the initial fracture permeability $10^{-3} \mu\text{m}^2$ and m is the permeability variation index. C_f and m can be obtained through laboratory testing or data fitting. Based on prior history-matching work, C_f values for both main and secondary fractures were set to 0.134 during the fracturing stage, with m values of 0.165 and 0.247, respectively; during the post-fracturing shut-in stage, C_f values were 0.016 and 0.033, with m values of 0.048 and 0.099, respectively. With the above reservoir and fluid properties defined, model initialization and dynamic simulation were completed.

Model Validation

Figure 3 shows the simulated distribution of fracture conductivity, indicating that the overall fracture network conductivity is low, with high-conductivity main fractures concentrated near the wellbore. The mid- to far-field is dominated by low-conductivity (blue) secondary fractures without proppant, consistent with hydraulic fracture support patterns reported internationally. Table 3 compares simulated fracture geometry with microseismic interpretation data. Due to differences in principles and accuracy of fracture characterization methods, microseismic results for different well sections exhibit a wide variation. The relative errors between simulated values and microseismic averages for half-length, width, and height are 7.6%, 4.2%, and 12.0%, respectively, and the simulated results fall within the corresponding microseismic value ranges, indicating that the numerical model can reasonably reflect the actual fracturing effect in horizontal wells. The example well experienced a 56-day shut-in period after fracturing; simulations of the fracturing injection and shut-in process, based on the fracture propagation model, further confirmed the applicability of the model.

Table 3—Comparison of simulated fracture dimensions with micro-seismic interpretation data

Fracture Dimension	Microseismic Data Range (m)	Microseismic Average (m)	Single-Stage Fracture Simulation (m)
Half-length	60–189	116.4	125.3
Width	50–140	93.6	89.7
Height	33–68	50.7	44.6

Case Study

An integrated numerical simulation approach was used to develop a series of shut-in models. First, the oil–water migration behavior between the hydraulic fracture network and reservoir matrix during shut-in was analyzed. Second, the influence of shut-in duration on post-fracture productivity and the optimization of shut-in time were investigated.

Oil–Water Migration Behavior

Based on the history-matched base model, shut-in was extended to 90 days. The dynamic distributions of pressure and water saturation within the fracture network during shut-in are shown in Figure 4. Most oil–water migration between media occurred during the early shut-in stage (first 30 days), characterized by a significant decline in pressure and water saturation within the fracture system, with corresponding energy increase in the formation. In the mid-to-late shut-in period (60 and 90 days), pressure and saturation further decreased but predominantly near the wellbore; changes slowed significantly, and far-field fractures stabilized. Analysis of fracture conductivity distribution (Figure 2) reveals a strong correlation between these spatial pressure–saturation differences and the distribution of main and secondary fractures. Near the wellbore, main fractures maintain larger apertures and higher net pressures at shut-in, whereas far-field secondary fractures have smaller apertures and lower net pressures, causing faster closure and earlier pressure equilibrium in the far field. Furthermore, under imbibition displacement during shut-in, water saturation in far-field secondary fractures declines more rapidly due to smaller aperture and lower retained fracturing fluid volume compared to main fractures, reaching oil–water displacement equilibrium earlier under similar matrix capillary pressures. Thus, oil–water migration during shut-in exhibits both temporal staging and strong spatial dependence on fracture complexity. The shut-in process can be divided into three main stages by time: (1) Fracture Closure Stage (days 1–5), where average fracture network pressure drops by 10.7 MPa and pore volume decreases by 9.0%, dominated by fracture closure. High fluid pressure inside fractures after shut-in drives leak-off into the matrix via pressurized displacement and imbibition, with an average pressure drop gradient of 2.1 MPa/day. (2) Imbibition Displacement Stage (days 6–50), where

pore volume further decreases by 10.5% but pressure drop gradient reduces to 0.082 MPa/day, fracture closure weakens, and fracturing fluid volume in the network decreases by 28.9%, dominated by matrix imbibition. (3) Energy Balance Stage (days 51–90), where pore volume and water volume decline rates synchronize, no additional oil enters fractures, and imbibition displacement reaches equilibrium. Although fractures continue to close slowly, their effect is minor (average pressure drop gradient 0.036 MPa/day), indicating energy balance between matrix and fracture systems within the stimulated region.



Figure 2—Core facies indicators of T1 oil-bearing formation in Lunnan

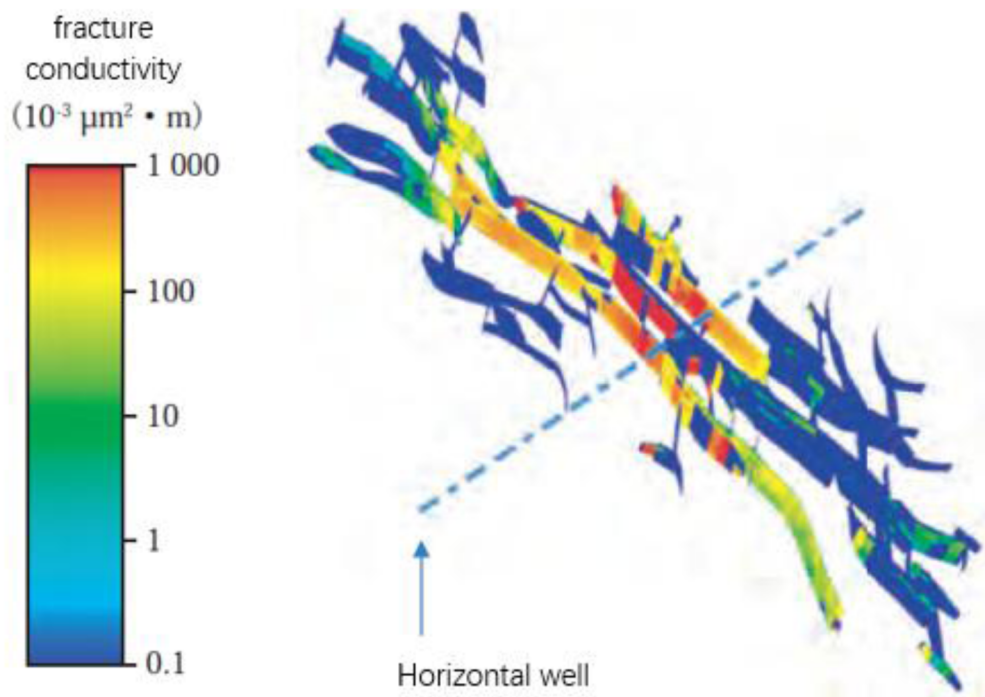


Figure 3—Distribution of fracture conductivity within the fracture network model

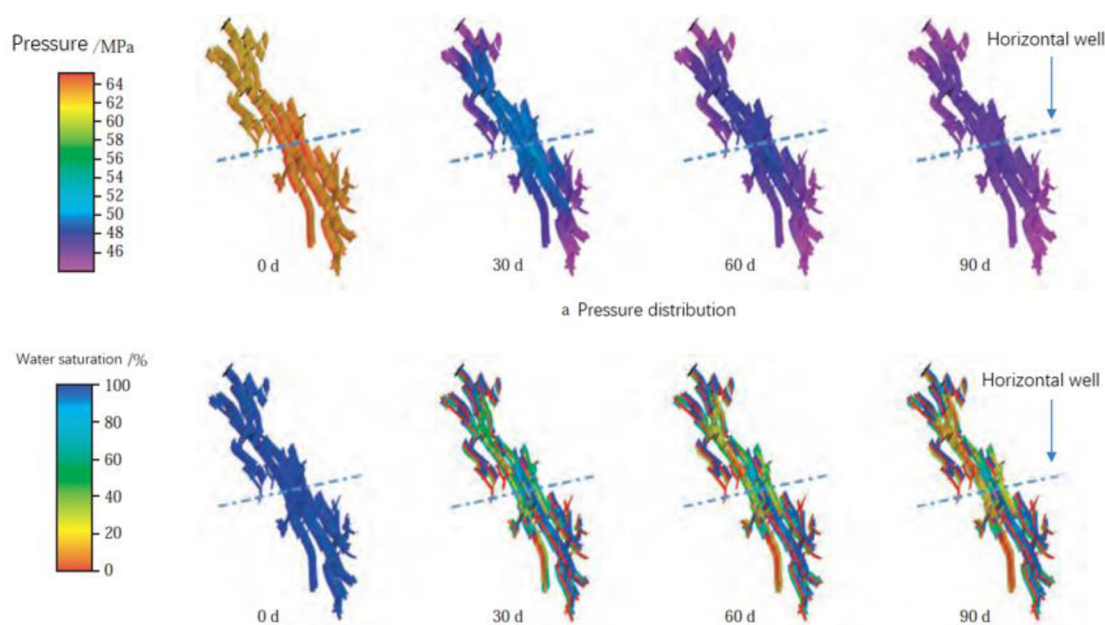


Figure 4—Simulated flow results within the fracture network during shut-in

Effect of Shut-In Duration on Productivity

To clarify the impact of shut-in duration on production dynamics of fractured shale oil horizontal wells, shut-in times of 1, 10, 20, 30, 45, 60, and 90 days were modeled. After shut-in, flowback was simulated for one month at a constant fluid rate of $7.5 \text{ m}^3/\text{day}$ (average daily production per fracture stage), followed by 11 months of production at a constant bottomhole pressure of 28 MPa. Production and flowback behaviors under different shut-in times are shown in Figure 5. Figure 6a shows that cumulative oil production increases with longer shut-in time, while fracturing fluid flowback rate decreases. This is due to matrix capillary imbibition during shut-in altering oil–water distribution within the fracture network; oil saturation in the network rises with longer shut-in, favoring early oil recovery. Simultaneously, more fracturing fluid imbibes into the shale matrix, where it is largely trapped and difficult to flow back, causing flowback rate reduction. Figure 6 compares incremental cumulative oil production relative to the 1-day shut-in baseline across different production durations and shut-in times. It reveals: (1) cumulative incremental oil production increases with shut-in time, but the incremental rate (per unit shut-in time) declines. For shut-in durations of 10, 20, and 30 days, incremental oil production rises nearly linearly with rates of 0.82, 0.93, and $1.06 \text{ m}^3/\text{day}$, respectively; for longer shut-ins of 45, 60, and 90 days, although cumulative increments still increase linearly, rates drop significantly to 0.29, 0.34, and $0.35 \text{ m}^3/\text{day}$, showing a clear inflection point. This is attributed to prolonged shut-in causing large volumes of fracturing fluid retention near fractures, creating high-water-content filtrate zones that weaken oil–water displacement between fractures and surrounding matrix. Additionally, reduced oil-phase relative permeability in these zones increases flow resistance during production. (2) For the same shut-in time, incremental oil production also increases with production duration. Since imbibition continues wherever fracturing fluid contacts the matrix, fracturing fluid within the near-fracture filtrate zone gradually migrates deeper into the matrix during production, partially alleviating initial relative permeability damage from high-water saturation. This results in incremental oil increases over production time, consistent with findings by Zhang Tao et al. The inflection point for shut-in benefit is observed consistently between 30 and 45 days, indicating that longer shut-in is not always better. It is recommended to begin production near the end of the imbibition-dominated stage to maximize oil–water displacement while minimizing fracturing fluid retention and controlling the extent of the filtrate zone.

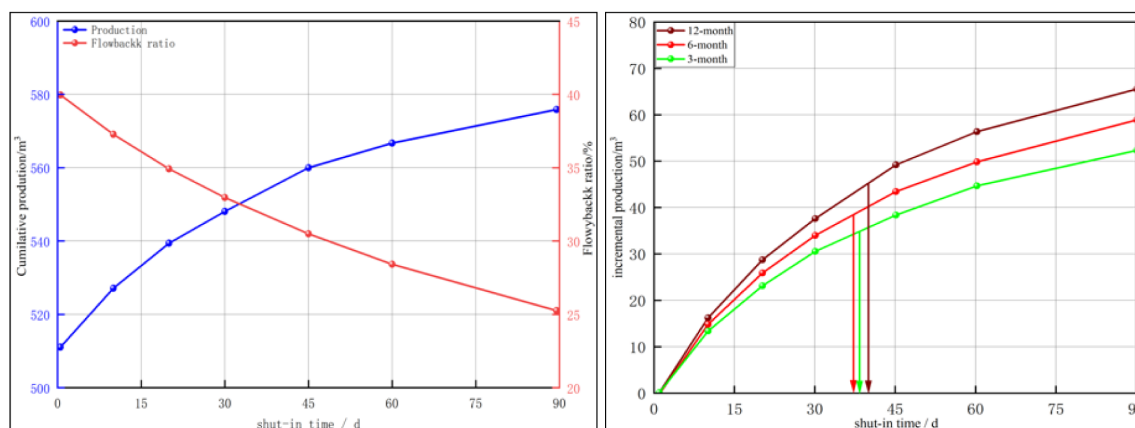


Figure 5—Production dynamics under different shut-in durations

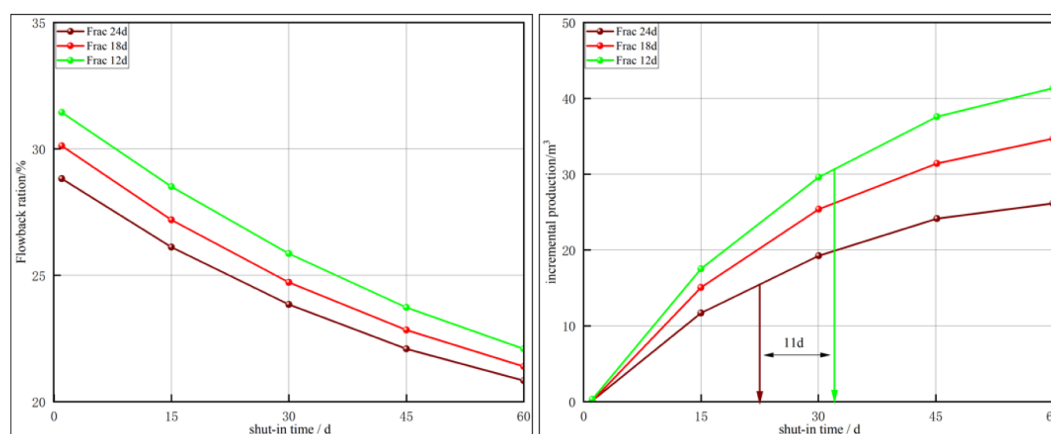


Figure 6—Production dynamics under different "fracturing + shut-in" durations

Impact of "Fracturing + Shut-In Time" on Productivity

Fracturing fluid contacts the matrix from the start of fracture propagation. Due to the long construction period of multistage fractured horizontal wells, the earlier fractured stages experience shut-in conditions similar to the overall well shut-in. Therefore, the shut-in duration varies non-uniformly along the wellbore. To further examine the effect of fracturing duration on optimal shut-in time, a multistage fracture model was developed with fracturing times of 12, 18, and 24 days; other geological and engineering parameters remained consistent with the base model. Production and flowback behaviors under different combined "fracturing + shut-in" durations are shown in Figure 7. Figure 7a shows that for the same shut-in time, flowback rate decreases further as fracturing duration increases. This indicates that fracturing fluid leak-off and displacement within previously fractured stages lead to increased formation fluid retention during fracturing. Figure 6 clearly shows that for a fixed shut-in time, cumulative incremental oil production over one year decreases as fracturing duration increases. Extended contact between fracturing fluid and shale does not result in significant production enhancement. For example, at 12 days fracturing, cumulative production growth noticeably slows after 45 days shut-in; at 24 days fracturing, incremental production plateaus after 30 days shut-in. The 11-day interval between these inflection points roughly corresponds to the 12-day difference in fracturing duration. Thus, for multistage fractured shale oil horizontal wells, optimizing shut-in time requires coordination with the fracturing schedule. In cases of operational delays or extended fracturing periods, reducing the originally planned optimal shut-in time by the delay duration can improve time efficiency and mitigate adverse effects of increased fracturing fluid retention.

Conclusions

From an integrated geo-engineering perspective, a numerical simulation method covering the entire fracturing–shut-in–production process was developed. It accurately characterizes the distribution of injected fracturing fluid within the fracture network and matrix system. Validation against microseismic monitoring and pressure history data confirms the method's applicability, providing an effective approach for optimizing reasonable shut-in time after volumetric fracturing.

Three main controlling stages of oil–water migration during shut-in were identified: fracture closure, imbibition displacement, and energy balance. The shut-in simulation results explain why extended shut-in favors early post-fracture oil production, while also revealing the time-limited nature of this effect. Prolonged shut-in leads to low fracturing fluid flowback, which may restrict long-term productivity, thereby clarifying the reasonable shut-in time under the combined effects of fracture closure, oil–water imbibition, and fracturing fluid retention.

Considering multistage fracturing in horizontal wells, the impact of fracturing duration on post-fracture production was clarified. Shut-in optimization must account for the fracturing schedule. For wells with operational anomalies or reduced fracturing efficiency, shut-in time can be appropriately shortened according to specific conditions, maximizing both time efficiency and shut-in benefits.

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